

Original research article

Simple and accurate calibration system for Laser Doppler Velocimeters

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ARTICLE INFO

Keywords:

Laser Doppler Velocimeter
 Laser Doppler Anemometer
 Calibration of Speed Measuring equipment

ABSTRACT

The calibration of Laser Doppler Velocimeters (LDVs) is of a great importance for the accurate measurement of the fluid flow rate and the speed of moving objects. In this paper, a simple, low-cost and portable calibration system for LDV is introduced. This system, which is based on a commercial optical chopper, can offer calibration uncertainty of (0.048%) ($\sigma = 2$) for velocities up to 1914 m/min and dynamic lengths of up to 15.4 km, which is better than the uncertainties reported by similar systems with a much complicated design. Therefore, this calibration system can be implemented easily by any calibration laboratory or even by national metrology institutes.

1. Introduction

The calibration of Laser Doppler Velocimeters (LDVs) is of a great importance for the accurate measurement of fluid flow rate and length/speed of moving objects. A simple, low-cost and portable method is required by small laboratories and even by National Metrology institutes (NMIs) to facilitate international comparisons. The spinning disk method is a well-established method for the calibration of LDV; moreover, it is considered the primary standard for LDV calibration [1]. Several techniques which are based on the spinning disk method have been introduced by some NMIs like: NMIJ (Japan) [2], NIST (USA) [3], PTB (Germany) [4], INMETRO (Brazil) [5] and INRIM (Italy) [6] as well as the US Naval surface warfare center (NSWC) [7]. The sandpaper technique, which was implemented at NSWC, uses a rotating sandpaper disk with a hole drilled precisely at the center of the disk, and the LDV laser is focused at accurately measured distance from center of the disk. In this technique, the distance should be measured precisely at every LDV calibration, which is considered a disadvantage. The spinning wire technique, which was implemented by NIST and NMIJ, uses a very thin wire (5 μm) to allow only single particle to pass through the LDV probe beam per revolution. The expanded uncertainty reported for this method is 0.47% for NIST and 0.2% for NMIJ [1]. This wire may be subject to deformation by the centrifugal forces by the rotating disk, which is considered again a disadvantage [6]. The Rotating glass disk technique, which is implemented by PTB, uses the edge of a precise rotating glass disk to simulate the gas flow. In this case, velocity is measured on the cylindrical surface normal to the axis of rotation rather than on the flat surface for the sandpaper disk. The PTB setup showed the best reported uncertainty over all other setups with expanded uncertainty of 0.055% [4], but the setup is fragile and heavy to transport. Both, INRIM and INMETRO, have used similar setup to the PTB but they used metallic disk instead of the PTB glass disk to enable the transportation. Although, researchers at INRIM have spent considerable efforts to make such system portable, the disk only weights 7.53 kg in addition to the large motor, alignment setup and controller which can reach 20 kg. Moreover, their reported uncertainty of 0.5% seems quite large. These systems are still far from being easily transportable internationally to facilitate international comparison. Moreover, they require attention during LDV adjustment to avoid wrong velocity measurement, despite the nice alignment

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system that has been made by INRIM.

In this paper, a simple system that based on a commercial optical chopper is introduced for the calibration of LDVs with expanded uncertainty of 0.048% ($\sigma = 2$), which is better than the best reported uncertainty so far. This system can be transported very easily since the optical chopper with its controller and the alignment holders has an overall weight of 3.8 kg (the chopper only weighs only 400 g, controller weighs 2.1 kg, alignment holder weighs 1.3 kg). In addition, this technique has other several advantages, since the small edge (0.3 mm) of the optical chopper blade helps to minimize the contributions to the uncertainty from the z-axis misalignment, the small thickness of the blade helps to avoid saturation of the high sensitivity detectors of Laser Doppler Anemometers (LDA) and its surface roughness is enough to scatter the LDV light for good detection SNR. The blade rotation speed is actively controlled with a lock-in amplifier to an internal oscillator up to speed of 1914 m/min (32 m/sec) and can be also locked to a high precision external oscillator to offer much more accurate rotation speeds. At the end of the paper, this system is implemented for the calibration of dynamic length measurement LDVs, since none of the previous publications tackled this issue.

2. Theory

LDV technique is based on detecting the Doppler frequency shift resulting from the scattering of light on a moving object [8]. Most of the LDV systems operate by splitting a laser beam into two equal intensity beams. Those two beams are made to intersect at angle (θ) from each other to form a set of fringes with spacing (δ) at the intersection volume. When an object crosses this volume perpendicular to the intersection, it scatters the light to a detector with frequency (f_{Doppler}) proportional to the speed of the object (V),

$$V = \delta f_{\text{Doppler}} \quad (1)$$

Calibration of LDV devices is performed by simulating the motion of an object by the rotation of a disk, whose velocity is described by [1]

$$V_{\text{ref}} = r\omega = 2\pi f_r r \quad (2)$$

Where, r is the disk radius, ω is the rotational speed in radians/s, and f_r is the rotational frequency in Hz (revolutions/s). When an optical chopper is used, a laser and a detector as well as a frequency counter can be used to detect the rotation frequency. For an optical chopper with 60 slits, the detected frequency should be divided by 60 to get the actual rotation frequency in Hz. For dynamic length calibration of LDVs, an oscilloscope is required to count the number of chopped pulses (N) from the spinning disk. The number of chopped pulses must be divided by 60 to get the actual number of rotations. Reference length is the length calculated from the number of pulses of the light passing the optical chopper in particular time frame. Therefore, the reference length can be represented by:

$$L_{\text{ref}} = 2\pi r (N/60) \quad (3)$$

By partially differentiating Eq. (2), the relative velocity deviation can be described by

$$\partial V/V = \sqrt{(\partial f/f)^2 + (\partial D/D)^2} \quad (4)$$

Where $D = 2r$ is the diameter of the disk.

3. System configuration

Fig. 1 shows the system used to calibrate LDV. An optical chopper from Thorlabs model (MC1000) with a blade that has 60 slit per revolution and thickness of 0.3 mm is used as the reference rotating disk. The disk diameter is measured accurately using profile projector 5 times at different position along the circumference. The average disk diameter is found to be 101.553 mm with uncertainty of $\pm 9 \mu\text{m}$. This measurement is traceable to the SI unit of length at NIS [9,10]. This chopper has a feedback circuit with integrated lock-in amplifier to control its rotation frequency to an internal oscillator. The rotation frequency is additionally measured with an external optical encoder that consists of a green He-Ne laser, a photodetector and a calibrated frequency counter to assure the measured rotation frequency. The number of slits per revolution has no effect on the accuracy of the rotation frequency measurement, since the frequency counter measures the frequency at the rising edge of the pulse. In other words, one revolution will start at an edge and ends at the same edge after one complete revolution. The LDV under calibration is a Laser-Speed 8000-3 from Beta-Laser-Mike

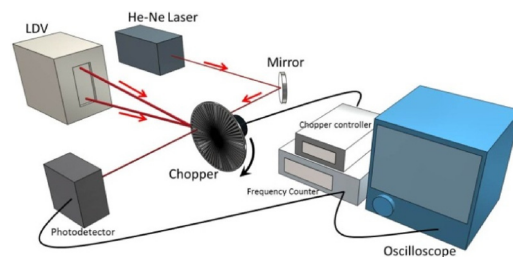


Fig. 1. LDV calibration setup.

with stand-off distance of 30 cm and depth-of-field of 35 mm and measurement rate of 100,000/s. The stated repeatability of the instrument is $\pm 0.02\%$ and the accuracy is $\pm 0.05\%$. This LDV is used by cable manufacturer for velocity and dynamic length measurement.

Due to the light weight of the optical chopper, it can be mounted easily on any optical mount to simplify the alignment and to lower the uncertainty, see Section 5. The small thickness of the disk helps to reduce the uncertainty due to the alignment of the LDV to disk. However, the detection SNR of the LDV may be reduced since the LDV beam diameter is around 1 mm while the disk edge is only 0.3 mm, but this SNR was sufficient for the measurement. The system calibrates LDVs by comparing its measured velocity with the simulated velocity by an optical chopper according to Eq. (2). Velocity calibration is sufficient for the LDVs that are used in anemometers. The dynamic length is the length of the rotating disk measured at one point at its circumference in particular time frame. For the calibration of dynamic length measuring LDVs, additional oscilloscope is needed to count the number of pulses. A shutter is used to enable and disable length measurement for LDV and the reference system at the same time. Thereafter, the reference length is calculated according to Eq. (3) and compared with the measured length by the LDV. Therefore, care must be taken to adjust the reference system beam exactly at the same level of the LDV beam; otherwise, mismatch between both measurements will take place.

4. Results

4.1. LDV calibration

In this section, the Laser-Speed 8000-3 is going to be calibrated for velocity and dynamic length measurement. Before starting the calibration process, the LDV must be aligned carefully such that the plane of the output laser from the LDV coincide with the plane of the optical chopper to avoid the errors described in Section 5. Some measurements are made also at single chopper velocity of 320 m/min to correct the parameters in the LDV software to minimize the velocity errors. Afterwards, the chopper is set at frequencies from 200 to 6000 Hz. From Eq. (2), these frequencies simulate velocities from around 60–1914 m/min. Calibration is performed by measuring the velocities displayed by LDV ($V_{LDV,i}$) at different reference chopper velocities ($V_{ref,i}$). The mean relative error is calculated for all velocities to be 0.001% with standard deviation of $u_{rep} = 0.01\%$. Fig. 2 represents a relation between reference velocities on the x-axis and the relative velocity error (difference between the measured and the reference velocities) on the y-axis.

A second-order polynomial regression is made on the curve above to obtain the behavior of the LDV in mathematical terms. This curve can be represented by the following equation: $2 \times 10^{-8} x^2 - 4.5 \times 10^{-5} x + 0.018$. It can be seen from Fig. 2 that some velocities have smaller relative errors than the others. However, the maximum variation over all velocities doesn't exceed the repeatability of the LDV measurement reported by the manufacturer which is 0.02%.

4.2. Thermal effects

This measurement is made at temperature of 25 °C. However, since the temperature variations greatly affect the calibration results. Additional set of measurements has been made at a single velocity for different temperatures to obtain the temperature correction coefficient of the LDV. This correction must be made if LDV operate at temperatures other than 25 °C. The measurements show change of 0.07 m/min per 1 °C (at a velocity of 319 m/min). Therefore the correction coefficient can be written as 0.022%/°C (Fig. 3).

4.3. Dynamic length calibration

For the calibration of the dynamic length of the same LDV, a shutter is used to start and to stop the measurement of the reference

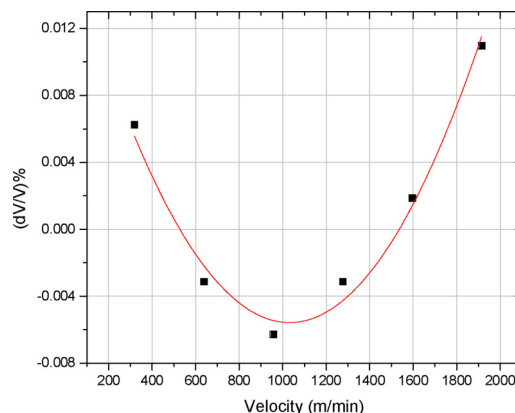


Fig. 2. Relative error in LDV for different velocities.

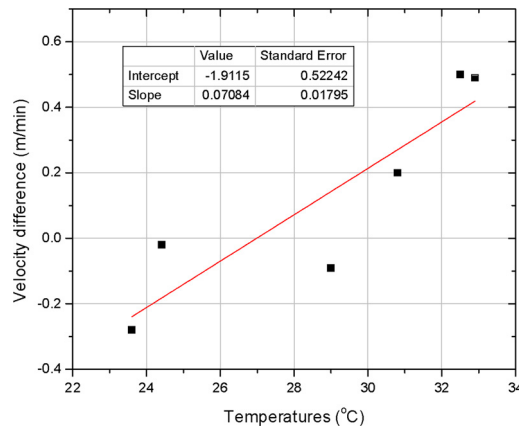


Fig. 3. Deviation of LDV reading with temperature variations.

system and of the LDV at the same time. The number of chopped pulses at different velocities is counted by an oscilloscope and converted to the reference length using Eq. (3). The length is calibrated for distances up to 15.4 km. The relative error of the LDV is plotted at each velocity in Fig. 4. The measurement is scattered randomly across the curve with average value of -0.01% and standard deviation of 0.01%.

4.4. Rotation frequency calibration

The optical chopper can be used alone for the calibration of LDV without using the external optical encoder (frequency counter, laser and photodetector). No degradation in the LDV calibration accuracy is expected since its internal encoder contains feedback circuit with lock-in amplifier to an internal quartz oscillator. In order to study its accuracy, the chopped light is measured with an external encoder system, while the optical chopper is adjusted at 1000 Hz. The frequency counter within the encoder is locked to the GPS disciplined oscillator to provide much accurate reference than the chopper internal encoder. After several measurements, the relative chopping frequency accuracy is found to be 6×10^{-6} , which corresponds to relative velocity accuracy of 0.0006%.

5. Uncertainty analysis

Since no measurement can be reported without uncertainty, uncertainty is calculated for LDV calibration results according to Eq. (4) and ISO 98-3 [11]. The main direct contributors to the uncertainty are: the disk diameter and its rotation frequency. However, there are other indirect contributors such as the thermal expansion of the rotating disk, the thermal expansion of the fringe spacing (8) and frequency counting in the LDV under investigation. In addition, the alignment of the LDV to the disk is crucial if not considered. In this section these parameters will be discussed.

5.1. The diameter of the spinning disk (u_D)

The diameter is measured with a profile projector that has accuracy of $1 \mu\text{m}$ at 5 positions around the circumference. The diameter

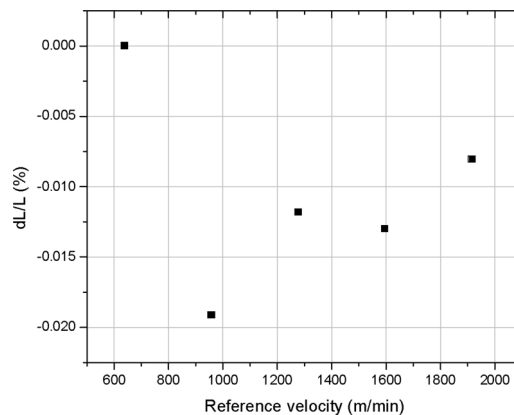


Fig. 4. Dynamic length calibration results.

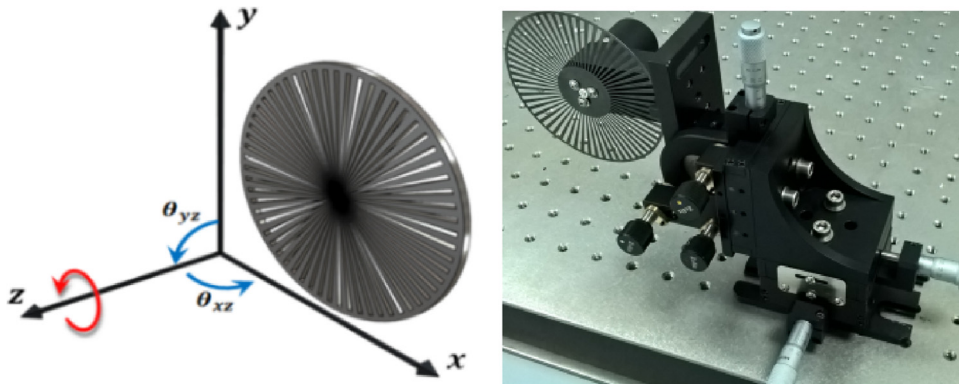


Fig. 5. Alignment of the rotating disk: (a) axes, (b) alignment stage.

is found to be 101.553 mm with standard deviation of 9 μm . Therefore, the uncertainty contribution from the diameter of the disk $u_D = 9 \mu\text{m} = 0.009\%$.

5.2. Rotation frequency (u_f)

In Section 4.4, a method is demonstrated for the calibration of the rotation frequency of the optical chopper with its internal frequency reference. This includes the uncertainty contributions from the internal oscillator, the frequency counter and the motor speed variations. The measured uncertainty of the rotation frequency is 6×10^{-6} which corresponds to a relative uncertainty of $u_f = 0.0006\%$.

5.3. Alignment (u_x, u_y, u_z)

The LDV laser beam must be aligned correctly to the edge of the spinning disk to assure accurate tangential velocity measurement. Any misalignment of the spinning disk in any of the three axes will lead to increasing the uncertainty; therefore, alignment must be taken into consideration. Fig. 5 shows (a) alignment axes, (b) the stage used for alignment.

5.3.1. X-axis alignment (Variation of fringe spacing along x-axis) (u_x)

From Eq. (1), the fringe spacing (8) of the LDV under calibration should be constant along the interference volume, otherwise, the calculation of velocity will be different for different positions in the interference volume. However, the fringe spacing is broader at edges than the center of the interference volume, as indicated in Fig. 6, which results lower velocities at the edges [12]. To overcome this problem, the maximum speed is searched along the LDV's depth-of-field (35 mm) at every calibration. Therefore, the uncertainty will lie within the repeatability of the LDV under calibration, which is $u_x = 0.01\%$ (the standard deviation from the results in Section 4.1) (Fig. 6).

5.3.2. Y-axis alignment (u_y)

The laser beam must be aligned to the center of the rotating disk to allow tangential velocity to be measured correctly and not a smaller component of this velocity. This source of uncertainty can be mostly eliminated during adjustment by shifting the disk up and down while searching the maximum velocity. Therefore, this error will be again within the repeatability of the LDV under calibration, which is $u_x = 0.01\%$ (the standard deviation from the results in Section 4.1) (Fig. 7).

5.3.3. Z-axis alignment (u_z)

the rotating disk must be aligned such that the plane of the disk is parallel to the plane of the LDV laser beam. Any tilt in the plane of the disk will cause a measurement error since a component of the disk velocity will be measured by the LDV. If the tilt angles are $(\theta_{xz}, \theta_{yz})$, the velocity component are $(V_{ref} \cos(\theta_{xz}), V_{ref} \cos(\theta_{yz}))$, see Fig. 8(a and b). The small thickness of the disk of 0.3 mm, which is smaller than the laser beam diameter of 1 mm, allows easy alignment of both angles without any helping tools. This has been done

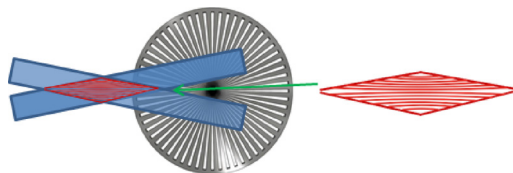


Fig. 6. Misalignment in the x-direction.

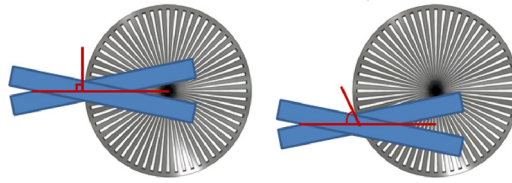


Fig. 7. Misalignment in the y-direction. Left: aligned, right: misaligned.

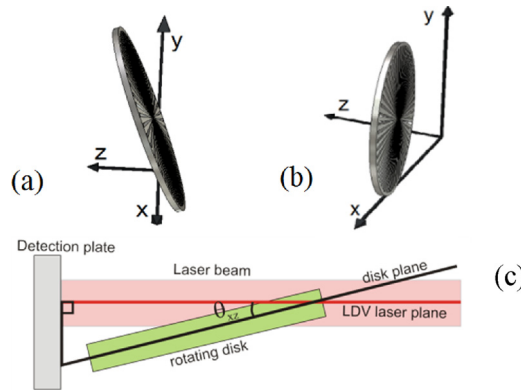


Fig. 8. Misalignment of the disk by: (a) an angle θ_{yz} (b) an angle θ_{xz} . (c) tilt adjustment using detection target.

by centering the disk in the middle of the beam at a detection target while moving the disk up and down in the y-direction to adjust both angles, see Fig. 8c. After alignment, the possible misalignment angle is estimated to be less than 0.6° , which lead to velocity error of $u_{\theta_{xz}}$, $u_{\theta_{yz}} = 0.0055\%$, and combined contribution from both angles to be $u_z = 0.0078\%$.

5.4. Temperature

Although our laboratory is thermally stabilized within 0.5°C , such temperature variation can affect the calibration results in two ways.

- (1) *Thermal expansion of the LDV (u_{TLDV}):* A change in the fringe spacing can result from the thermal expansion of the LDV under test. The LDV sensitivity to temperature has been studied experimentally in Section 4.2. For each LDV under calibration such correction factor must be found. However, an uncertainty in laboratory temperature of 0.5°C can lead to a relative velocity uncertainty of 0.011% , which is calculated from sensitivity factor given in Section 4.2 ($0.022\%/^\circ\text{C}$).
- (2) *Thermal expansion of the disk (u_{TD}):* A temperature uncertainty of 0.5°C can cause uncertainty in the diameter of disk, which is made from steel, due to its thermal expansion by 0.0005% .

5.5. Other sources related to LDV

There are some other uncertainty sources from the LDV instrument itself, such as its internal burst spectrum analyzer. Although these sources of uncertainty have nothing to do with our calibration system, these contributions are included in the repeatability of the LDV instrument under calibration (u_{rep}).

5.6. Leveling (u_L) (for dynamic length calibration only)

For dynamic length calibration, both LDV and encoder beams should be at the same level. If an estimated difference of 1 mm exists, an error in the reference length of 0.001% will present (assuming a speed of 318 m/min or 1000 pulse/second , 100 s acquisition time and shutter speed of 1 m/second). Since the level difference between the two beams could adjusted easily to 1 mm or even less, this uncertainty source can be neglected.

6. Combined uncertainty

Table 1 summarizes the contributions to the uncertainty that has been discussed earlier in this section.

The combined uncertainty in speed calibration can be calculated by adding in quadrature the sources of uncertainty in Table 1:

$$u_c = \sqrt{u_D^2 + u_f^2 + u_x^2 + u_y^2 + u_z^2 + u_{TLDV}^2 + u_{TD}^2 + u_{rep}^2} = 0.024\% \quad (5)$$

Table 1
Uncertainty budget for LDV calibration system.

Parameter	Value	Speed uncertainty (%)
Diameter of the disk (Type B)	9 μm	0.009 %
Rotation frequency (Type A)	6×10^{-6}	0.0006%
x-axis alignment (Type A) (δ variation along x-axis)	0.01 %	0.01 %
y-axis alignment (Type A)	0.01 %	0.01 %
z-axis alignment (Type B)	0.6°	0.0078%.
Thermal expansion of LDV (Type A)	0.5 °C $\times$ 0.022%/°C.	0.011%
Thermal expansion of the disk (Type B)	10 $\mu\text{m}/^\circ\text{C} \times 101 \text{ mm} \times 0.5 ^\circ\text{C}$	0.0005%
Repeatability (Type A)	0.01%	0.01%
Combined uncertainty		0.024%
Expanded uncertainty ($\sigma = 2$)		0.048%

The expanded uncertainty is calculated to be $U = 0.048\%$ for coverage factor of ($\sigma = 2$). Although the system is simple, this calibration uncertainty is better than the uncertainties reported for similar systems so far, which was the result of decreasing the uncertainties from alignment of the disk. For dynamic length calibration, the contribution of levelling of $u_L = 0.001\%$ is added to Eq. (5), which will give almost the same uncertainty as in velocity.

7. Conclusion

A system based on a commercial optical chopper is developed for the calibration of laser Doppler velocimeters (LDV) with uncertainty better than 0.048% ($\sigma = 2$) for velocities up to 1914 m/min, which is better than the reported uncertainties for similar systems by other metrology institutes. This system is implemented for the calibration of a commercial LDV device for velocities up to 1914 m/min and dynamic lengths of 15.4 km. The average error is found to be 0.001% for velocity and -0.010% for length with expanded uncertainty of 0.048%, which is limited by the LDV under calibration.

References

- [1] ITTC, Uncertainty Analysis - Laser Doppler Velocimetry Calibration, ITTC Procedure, (2008).
- [2] N. Kurihara, Y. Terao, S. Nakao, et al., Development and Uncertainty Analysis of a Laser Doppler Velocimeter, Trans. Jpn. Soc. Mech. Eng. Ser. B 65 (637) (1999) 3029–3034.
- [3] T.T. Yeh, J.M. Hall, An uncertainty analysis of the NIST airspeed standards, 5th Joint ASME/JSME Fluids Engineering Conference (2007).
- [4] V. Strunck, N. Pape, H. Müller, et al., Measurement uncertainty of the realisation of velocity for the calibration of LDA with a rotating disk, 18th Fachtagung, Lasermethoden in der Strömungsmesstechnik, (2010).
- [5] F.O. Costa, F.S. Ferreira, J.L.Z. Zotin, et al., Calibration of a laser-doppler anemometer by means of calibration disk, 22nd International Congress of Mechanical Engineering (2013).
- [6] A. Piccato, C. Francese, R. Malvano, A portable rotating disk prototype for LDA calibration, Flow Meas. Instrum. 38 (54) (2014) 61.
- [7] J.T. Park, J.M. Cutbirth, W.H. Brewer, Hydrodynamic Performance of the Large Cavitation Channel (LCC), Technical Report NSWCDD-50-TR—2002/068, Naval Surface Warfare Center, USA, 2002.
- [8] Zh Zhang, LDA Application Methods—Laser Doppler Anemometry for Fluid Dynamics, Springer, 2010.
- [9] L. Robertsson, M. Zucco, L.-S. Ma, et al., Results from the CI-2004 campaign at the BIPM of the BIPM.L-K11 ongoing key comparison, Metrologia 42 (1) (2005) 22.
- [10] O. Terra, H. Hussein, An ultra-stable optical frequency standard for telecommunication purposes based upon the $5S_{1/2} \rightarrow 5D_{5/2}$ two-photon transition in rubidium, Appl. Phys. B 122 (2016) 27.
- [11] ISO 98-3, A Guide to the Expression of Uncertainty in Measurement, (2008).
- [12] Z. Zhang, LDA Application Methods—Laser Doppler Anemometry for Fluid Dynamics, Springer, 2010.